Structural certificates for the Suzuki and Janko small-core tables

Private L054 finite application

Theorem 1. *With the acquired numerical inventories and named-model identifications, all 247 current companion rows have their displayed structural descriptions: 40 for J_1, 146 for J_2, 22 for* Sz(8) *and 39 for* Sz(8) : 3*. An extension label asserts existence of its stated decomposition, not uniqueness of a group defined by the label.*

Sources and exact embeddings. All eleven native pages of the current Connor–Leemans companion PDFs were directly viewed. Powers and dots are preserved. The published left-association convention gives $(3 \cdot A_6) : 2$ in J_2 row 3 and $3 \cdot A_6$ in row 6. Row 13 retains the literal source $2^{1+4} : (D_{10})$; its redundant parentheses around the single atom are recorded, while the semantic tree has the same D_{10} factor. Parentheses around composite factors are not removed.

The six-table numerical component and its distinct root review are acquired. Its actual literal final states, complete row bijections and named-model checks supply the present occurrences. The states contain explicit embedded generators, on 266 points for J_1, 100 for J_2 and 65 for Suzuki. No displayed constructor name is substituted for a proper embedding.

The retained J_1 model is primitive-library entry $(266, 1)$, and J_2 is the derived subgroup of entry $(100, 2)$; their identifications are acquired from the accepted sporadic representation and six-table interfaces. The two top-row generating lists are literally equal to these already identified final-state lists. Their naming certificates are therefore exact identity joins, not fresh large Cayley-table computations. The Suzuki model is the accepted named ovoid action, with the explicit coordinate square map for its order-three field extension. These identification premises are retained explicitly; matching orders alone are not the naming argument.

Structural reuse. The registry uses 142 earlier structural models and three named inputs. There are 33 new decomposition models: 25 split extensions, six nonsplit extensions, one direct product and one extraspecial model. Every proper occurrence and factor map is a literal bijection preserving identity and every right-generator edge. The acquired MC.3 word-induction argument proves each map is an isomorphism. Normality is checked by conjugating factor generators by group generators. An actual complement has trivial intersection and the required product of orders. Direct factors commute; a displayed nontrivial semidirect action has noncommuting generators. The acquired prime, cyclic-six, S_3, Klein-four and extraspecial criteria are used without alteration. The only new nonsplitting argument follows.

Lemma 2 (Central commuting-lifts obstruction)**.** *Let $N \leq Z(H)$ and let $x, y \in H$ have nonidentity commutator $[x, y] \in N$. Then the extension of H/N by N has no complement.*

Proof. The images of x, y commute in H/N. Every pair of their lifts has the form nx, my with $n, m \in N$. Centrality gives $[nx, my] = [x, y] \neq 1$. If a complement existed, the inverse of its quotient isomorphism would give commuting lifts of these two commuting elements, a contradiction. □

The actual $3 \cdot A_6$ certificate. Inside the saved J_2 row-6 group of order 1080, an actual normal subgroup N of order three is identified with the cyclic model. The reader checks that every element of N commutes with every group generator, hence $N \leq Z(H)$. The proposed x, y belong to H, and their literal commutator is a nonidentity element of N. All nine shifted commutators $[nx, my]$ are also checked directly.

The quotient name is paid separately. The certificate gives 360 actual representatives of right cosets Nr. The reader forms every coset, verifies their disjoint complete union, and computes the right action of every ordered generator on these cosets. Since N is normal, this action has kernel exactly N. Its literal image has order 360. An explicit bijection from the acquired A_6 model to that image preserves all generator edges, proving $H/N \cong A_6$. Thus the lemma proves exactly the source dot in $3 \cdot A_6$. The row-3 group has a separate actual index-two complement, proving $(3 \cdot A_6) : 2$.

Finite evidence and development. The first bounded witness pass returned in 11.41 seconds with 245 successful rows and two explicit failures, J_2 rows 3 and 6. A proposed single-element order obstruction did not certify those rows. It is not a premise or an accepted result. Root then supplied the central commuting-lifts argument. A changed two-row pass completed in 1.77 seconds, reusing all earlier witnesses and producing the missing two certificates. Both executions and the exact earlier code are retained. No ambient subgroup census was repeated.

Ordinary and optimized replay use literal permutations and must agree. Small quaternion controls validate the central obstruction and reject a noncentral kernel, a trivial commutator, missing or duplicate quotient cosets. Separate source, named-identity and embedding damages are rejected. The reader also handles GAP's physical numeric line continuation before a comma; nonnumeric continuations remain invalid. This concerns serialization, not a mathematical change. The initial source-parser rejection of redundant atomic parentheses is preserved; raw source labels remain exact.

The first identity check rejected a GAP-proposed automorphism where literal generator equality was required. The successor writes the unchanged generators explicitly; both editions and the failed receipt are preserved. This application claims current-table structural joins only. It does not claim historical PDF byte identity, canonical extension labels, whole-source admission, new question resolution, formalization or measurement. `1044_resume` authored the application and literal certificates; root contributed the commuting-lifts argument. A different agent conducts the distinct mathematical review.